PRASHEEL NANDWANA

prasheelnandwana@gmail.com | prasheel.com | linkedin.com/in/withprasheel/ | github.com/withPrasheel

Education

Arizona State University **August 2023 - May 2025**
Master of Science in Software Engineering (GPA: 4/4) *Tempe, United States*
Relevant Courses: *Advanced Data Structures & Algorithms, Cloud Computing, Software Validation and Testing, Web-Based Applications*

PES University **July 2016 - August 2020**
Bachelor of Engineering in Electronics and Communication Engineering *Bangalore, India*

Professional Experience

ASM **July 2025 - Present**
Software Engineer II *Phoenix, United States*

- Work with the Epitaxy team to design and implement advanced features for proprietary software platforms, distributed to both vendors and clients, using C++ and Object Pascal.
- Develop and maintain hardware communication drivers that enhance real-time data exchange between the control system and epitaxial equipment, reducing latency and improving response consistency.
- Build and refine scheduler logic to synchronize parallel reactor operations, improving throughput and system reliability.
- Spearheading the development of enhanced capabilities in the scheduler system, improving real-time communication and synchronization between critical manufacturing equipment.
- Collaborating with cross-functional teams to debug, test, and optimize system components, ensuring compliance with internal standards and manufacturing tool specifications.

Netmeds (Reliance Group) **December 2021 - July 2023**
Software Development Engineer *Chennai, India*

- Implemented a scalable **microservices** architecture using advanced **OOP Paradigms**, **Java-Spring Boot**, and **Angular** and **TypeScript** features to efficiently manage over 100k e-commerce orders per day as a **Full Stack Developer.**
- Developed robust **RESTful APIs** for **CRUD** operations, with rigorous unit testing with **JUnit 5** to ensure system reliability
- Secured authentication flows using OAuth 2.0, JWT tokens, and implemented RBAC-based API security.
- Designed and managed SQL database indexing and partitioning, improving query performance by 3x.
- Designed adaptable architectures for critical components using optimal **design patterns** impacting 100k+ customers.
- Integrated SAP for ledger tracking, automating financial reconciliation using asynchronous event-driven pipelines (Kafka, RabbitMQ).
- Incorporated a **BDD pipeline** leveraging **Cucumber** and **Java**, reducing manual effort by 99.46%, slashing time from 16 hours to 6 minutes.
- Enhanced **system security** by effectively managing and addressing security reports, successfully resolving a total of 19 individual security bugs.
- Led and mentored a team of 3 to write automated test cases, showcasing leadership and management skills.

Synechron **October 2020 - November 2021**
Software Engineer *Bangalore, India*

- Contributed to **migrating** a Legacy system to Java, demonstrating proficiency in navigating intricate processes and seamlessly transitioning between Cobol and Java.
- Designed an intuitive web application for streamlined payroll processing, employing **Angular** and **Node.js** technologies to benefit 1000+ employees, ensuring scalability and modularity.
- Implemented **event-driven payroll processing using Kafka**, ensuring fault-tolerant salary calculations and tax deductions .
- Designed and deployed a scalable payroll application on Kubernetes (EKS), optimizing cost and performance by autoscaling based on pay cycle peaks.
- Well-versed with **agile** methodology, version control systems, customer interaction, and troubleshooting.

Relevant Projects

GitHub Ideal Behaviours **September 2024 - November 2024**

- Crafted a Python-based system using **FastAPI** framework to automatically categorize developers by their Git contributions, resulting in over 60% reduction in managerial investigation time.
- Employed a robust **MVC architecture** with Pydantic models to optimize data handling and system structure, achieving a code coverage of 96% through **Pytest unit testing**

Elastic Face Recognition on AWS **June 2024 - August 2024**

- Designed a scalable facial recognition application using **FastAPI**, leveraging AWS web services like **EC2 and S3** to handle the variable workloads.
- Orchestrated asynchronous communication between tiers via **Kafka**, efficiently processing up to 100 requests within 3 minutes.
- Developed a custom auto-scaling algorithm to dynamically adjust EC2 instances based on demand.

Care Chat **May 2024 - June 2024**

- Developed a full-stack real-time conversational chatbot with **React** (frontend), **Express-Node.js** (backend), and **MySQL** (database), integrating with **OpenAI LLM API**.
- Implemented **JWT token** authentication for secure login and data transactions and utilized Gmail **SMTP** for sending verification links, ensuring reliable conversation history storage in MySQL.

Technical Skills

Languages: Java, C++, JavaScript, Python, SQL, HTML/CSS, TypeScript, Bash
Backend & Distributed Systems: Spring Boot, FastAPI, J2EE, REST APIs, NodeJs, Database Sharding, Kafka
Cloud & Devops: AWS (Certified Solutions Architect – Associate), EC2, S3, Lambda, RDS, VPC, Load Balancing, Auto Scaling, Docker, Kubernetes, Terraform, CI/CD (Jenkins, GitHub Actions, AWS CodePipeline)
Databases: MySQL, PostgreSQL, SQL Server, Oracle, DerbyDB, MongoDB, DynamoDB, Cassandra
Frontend & UI Development : Angular, ReactJS, VueJS, JWT, OAuth2, Secure API Integration
Testing & Quality Assurance: JUnit, Cucumber, Postman, Selenium
Other: Microservices, Integration, Designing, Planning, Development & Deployment, Design Patterns, Debugging, Unit Testing, Root Cause Analysis (RCA), Software Development Life Cycle (SDLC), Excel, Linux, Unix, Agile, Scrum, Jira